

FREESTANDING BENCH · 64 X 24-1/2 IN · 5 IN CLEAR FLOOR

Freestanding Bench Cut List

The client's change of 2026-10-26: the bench is now a freestanding piece with 5 in of clear floor under it for the robot vacuum. The alcove's baseboard stays, the bench slides in with 1 in of air to each wall and its back 1/2 in off the wall. Kept from the built-in: all-white frame-and-panel fronts with brass pulls, the solid maple top at 14 in (seat 17 in with the cushion), two drawers on 21 in Blum 563H undermounts, 1/2 in Baltic birch boxes, Osmo Natural on the bare maple.

Changed: a 64 x 24-1/2 in top over a 62-1/2 in case, the case ends are show faces (painted white, edges banded), four 2 in square maple legs on a C-frame of 3/4 x 2-1/2 in maple rails with no front rail, drawers 6-1/2 in tall instead of 8-1/2 (the 5 in of floor come out of the drawers), no plinth, no scribe strips, no wall screws. The base style (legs, blades or gables) is Brian's pick on the base options page; this page documents the legs. The built-in design stays on its own shop page in case the client goes back.

34 pieces: the case first, then the drawers, each grouped by material in cutting order. Inches to the nearest 1/32. Every part has a shop drawing with its grooves, rabbets, notches and bores located from the blank's own edges.



NUMBERS THAT MOVED

- **Clear floor under the case** 5 in (the tallest current Roomba is under 4 in)
- **Drawer box** 6-1/2 in tall, 5-17/32 in inside (was 8-1/2 in)
- **Drawer fronts** 31-1/8 x 8-1/8 in (were 31-3/32 x 10-1/8)
- **Top** 64 x 24-1/2 in, 1 in of air to each wall (was 66 x 24-5/8, wall to wall)
- **Case** 62-1/2 x 23 x 8-1/4 in, back 1/2 in off the wall
- **Legs** 2 x 2 x 5 in, outer faces 2-1/2 in from the case ends, 2 in behind the case front edge and 2 in inside the case back
- **Overall** 64 x 24-1/2 x 17 in with the cushion; wood top at 14 in, seat 17 in, unchanged

Measure the alcove at the floor, 16 in and 36 in, at the back wall and the pilaster faces, before cutting. The piece is 64 in wide; if the opening is under 66 in anywhere, narrow the top. The baseboard stays: the back legs sit 2 in inside the case back, so confirm the baseboard and shoe stand no more than 1-1/4 in off the wall. The low receptacle seen in the photo ends up behind the case; nothing can stay plugged into it.

Case 14 pieces

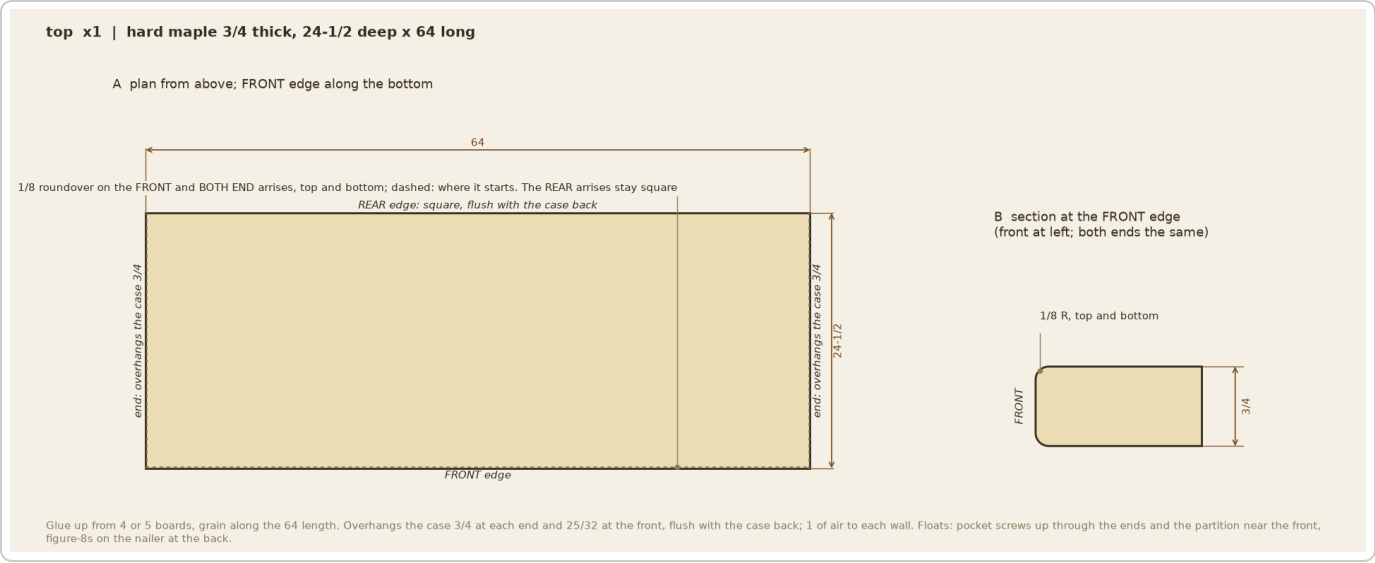
Hard maple: top, nailer, legs and base rails (bare, Osmo Natural)

9 pieces · 13.7 sq ft face, no kerf or waste

1× top

3/4 x 24-1/2 x 64

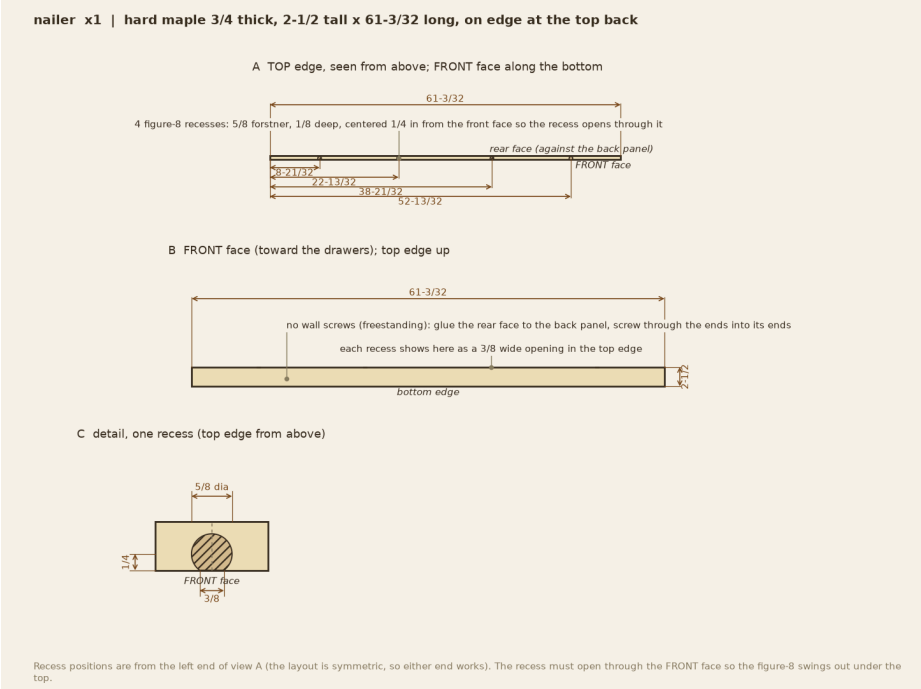
glue up from 4 or 5 boards, grain along the 64; 1/8 roundover on the front and both end arrises, top and bottom (the back edge stays square against the wall); overhangs the case 3/4 at each end and 25/32 at the front, flush with the case back; pocket screws up through the ends and partition near the front, figure-8s on the nailer at the back



1× **nailer**

3/4 x 2-1/2 x 61-3/32

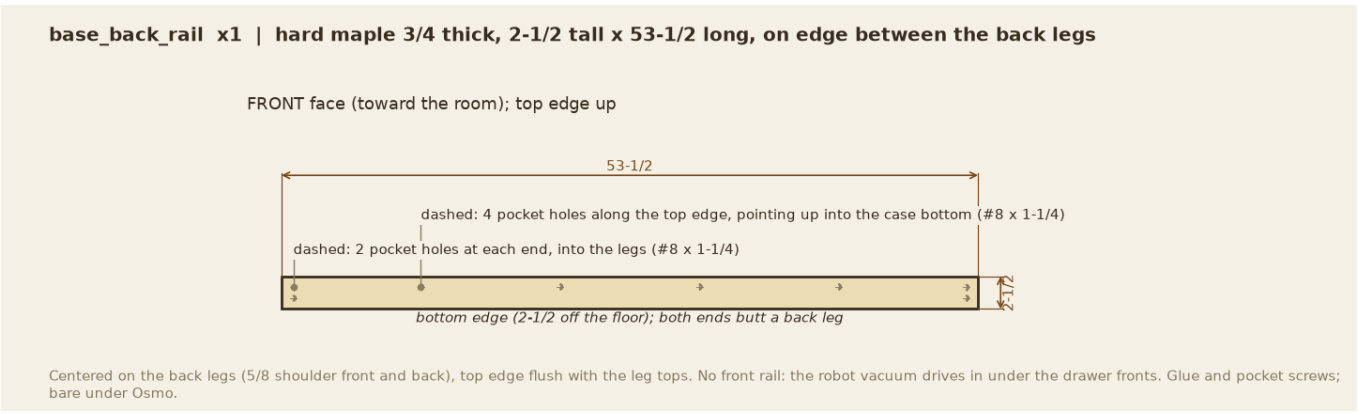
on edge against the back panel, through the partition notch; four 5/8 figure-8 recesses 1/8 deep on the top edge, centered 1/4 in from the front face; glued to the back, screwed through the ends into its ends (no wall screws: the piece is freestanding)



1× **base_back_rail**

3/4 x 2-1/2 x 53-1/2

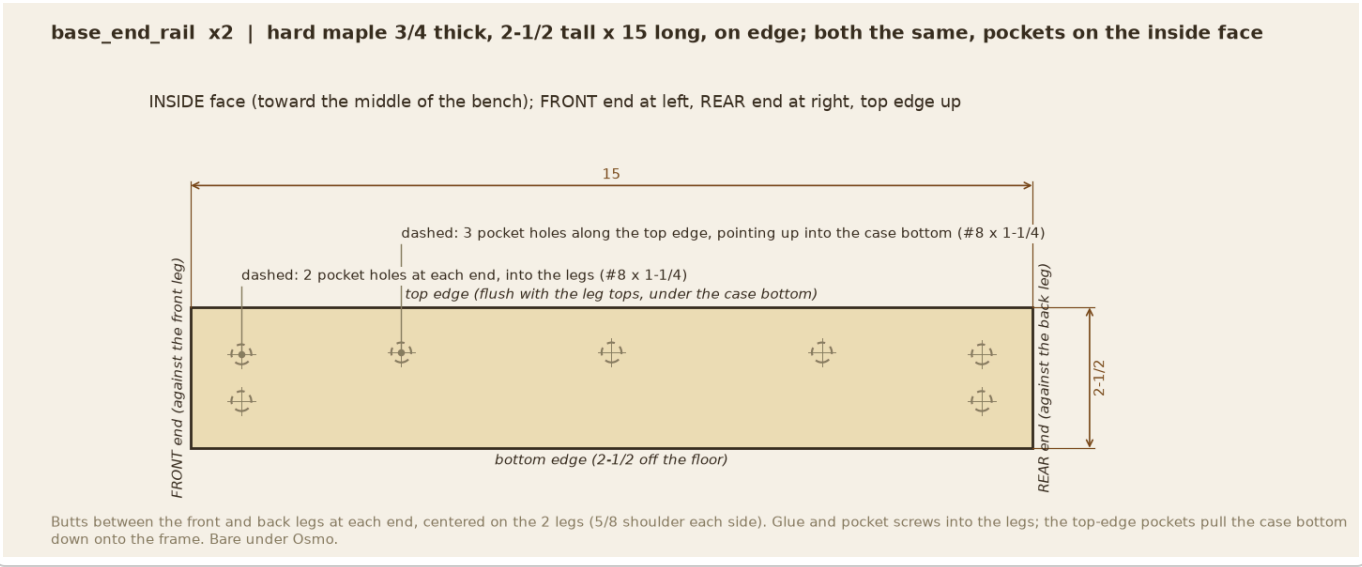
on edge between the two back legs, centered on them, top edge flush with the leg tops; glue and two pocket screws into each leg from the front face; four pocket screws up into the case bottom; no front rail, the robot vacuum drives in under the fronts



2× base_end_rail

3/4 x 2-1/2 x 15

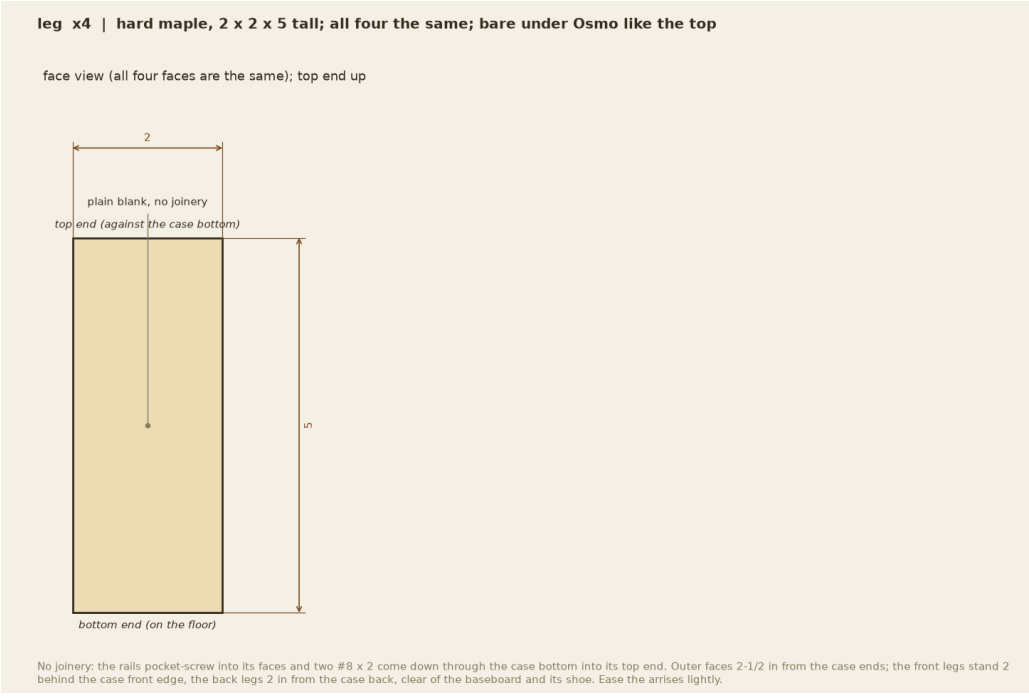
on edge between the front and back legs, centered on them, top edge flush with the leg tops; glue and two pocket screws into each leg from the inside face; three pocket screws up into the case bottom



4× leg

2 x 2 x 5

2 square, 5 tall; bare maple under Osmo like the top; glued and screwed to the case bottom with two #8 x 2 down through it (under the drawer boxes), rails pocket-screwed into them



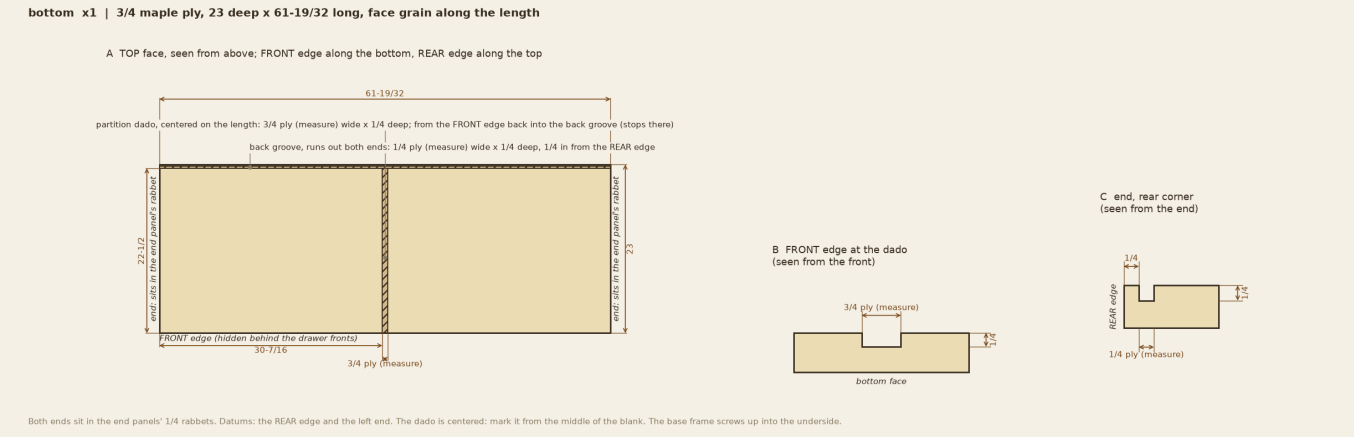
3/4 in maple plywood (measure the sheet): ends (show faces, painted white, edges banded), bottom, partition

4 pieces · 13.7 sq ft face, no kerf or waste

1× bottom

23/32 x 23 x 61-19/32

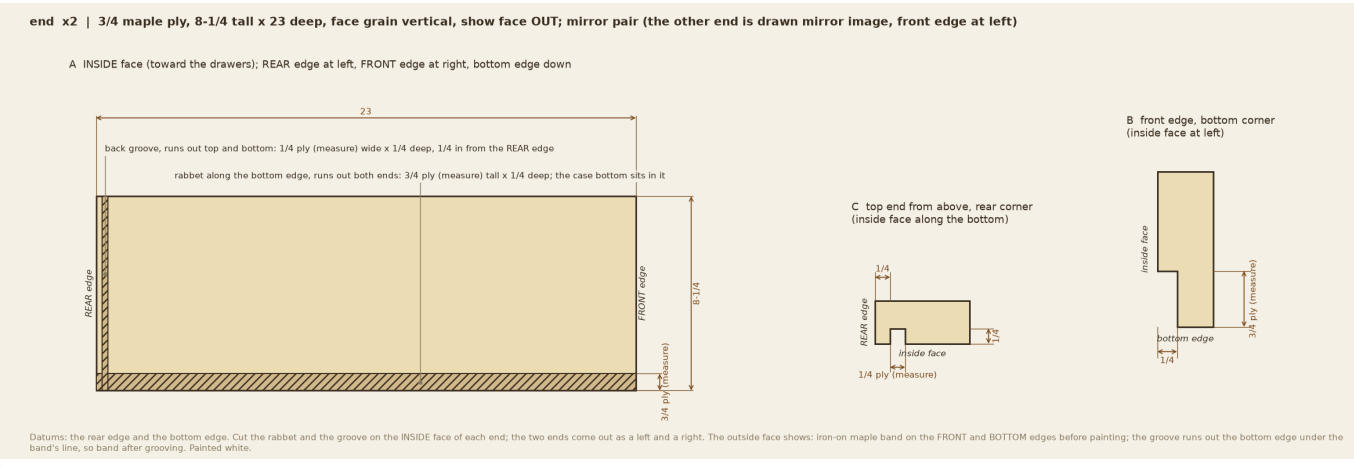
face grain across the 62-1/2; 1/4 deep back groove 1/4 from the rear edge; 1/4 deep x 3/4 ply (measure) dado for the partition, centered, from the back groove to the front edge; front edge is covered by the drawer fronts; the base screws to its underside



2× end see drawing: not a plain rectangle

23/32 x 8-1/4 x 23

show face out, face grain vertical; 1/4 x 23/32 (measure) rabbet along the bottom inside edge for the case bottom; 1/4 deep x 1/4 ply (measure) groove 1/4 from the rear edge, through; iron-on maple band on the front and bottom edges; painted white; mirror pair



1× **partition**

23/32 x 7-25/32 x 22-1/2

face grain vertical; sits in the bottom dado and against the back panel; notch the top rear corner 3/4 deep x 2-1/2 tall for the nailer



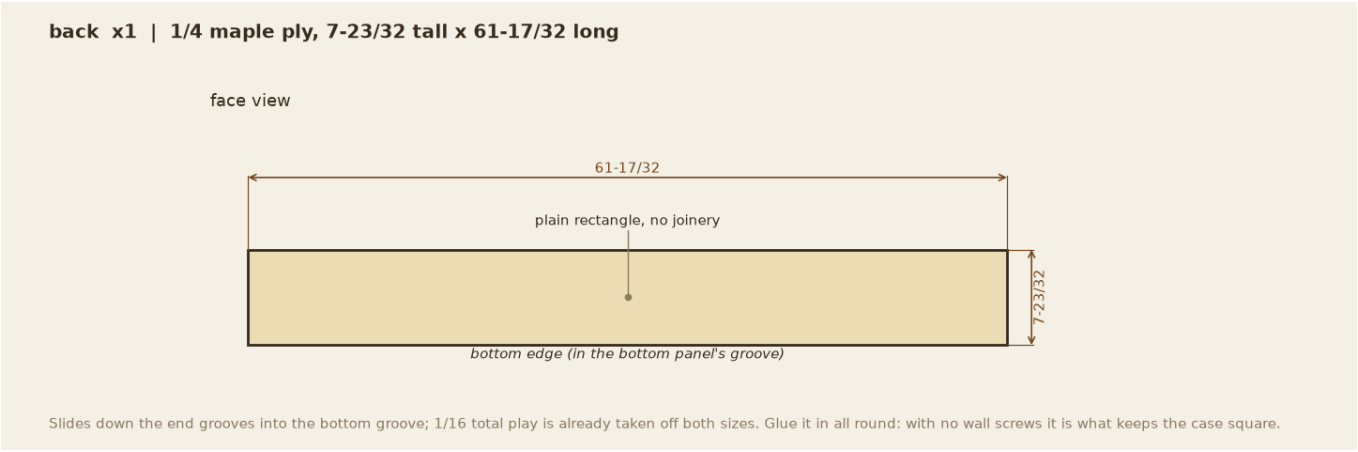
1/4 in maple plywood (measure the sheet): back

1 pieces · 3.3 sq ft face, no kerf or waste

1× **back**

1/4 x 7-23/32 x 61-17/32

plain rectangle; slides down the end grooves into the bottom groove; glued in all round (with no wall screws it is what keeps the case square)



Drawers 20 pieces

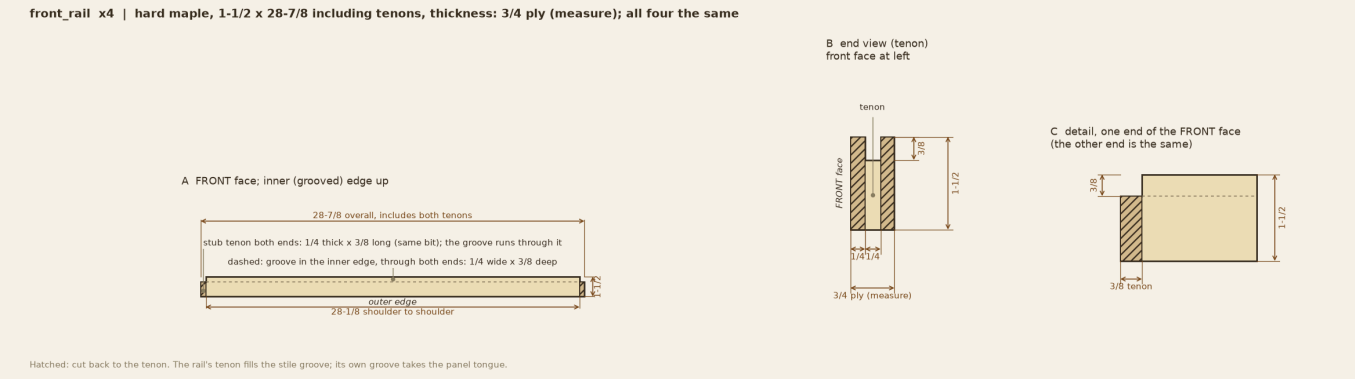
Drawer front frames, hard maple, painted white: stiles and rails

8 pieces · 1.5 sq ft face, no kerf or waste

4× front_rail see drawing: not a plain rectangle

23/32 x 1-1/2 x 28-7/8

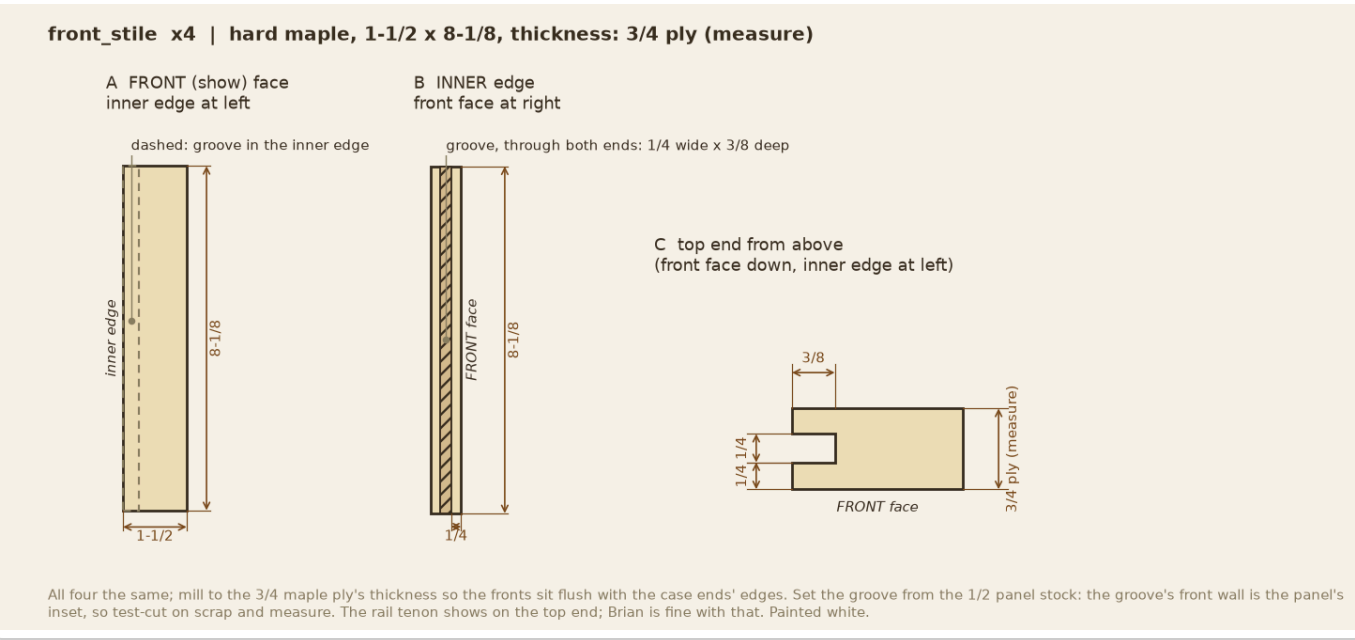
1/4 x 3/8 groove on the inner edge, through; 1/4 x 3/8 stub tenon on both ends (same bit); length includes both tenons; painted white



4× front_stile see drawing: not a plain rectangle

23/32 x 1-1/2 x 8-1/8

1/4 x 3/8 groove on the inner edge, through, front wall 1/4 thick (set the fence from the panel ply); the rail tenons show on both ends (the top one is the visible one); painted white



Drawer front panels, 1/2 in MDF or paint-grade plywood (measure the sheet), painted white with the frames

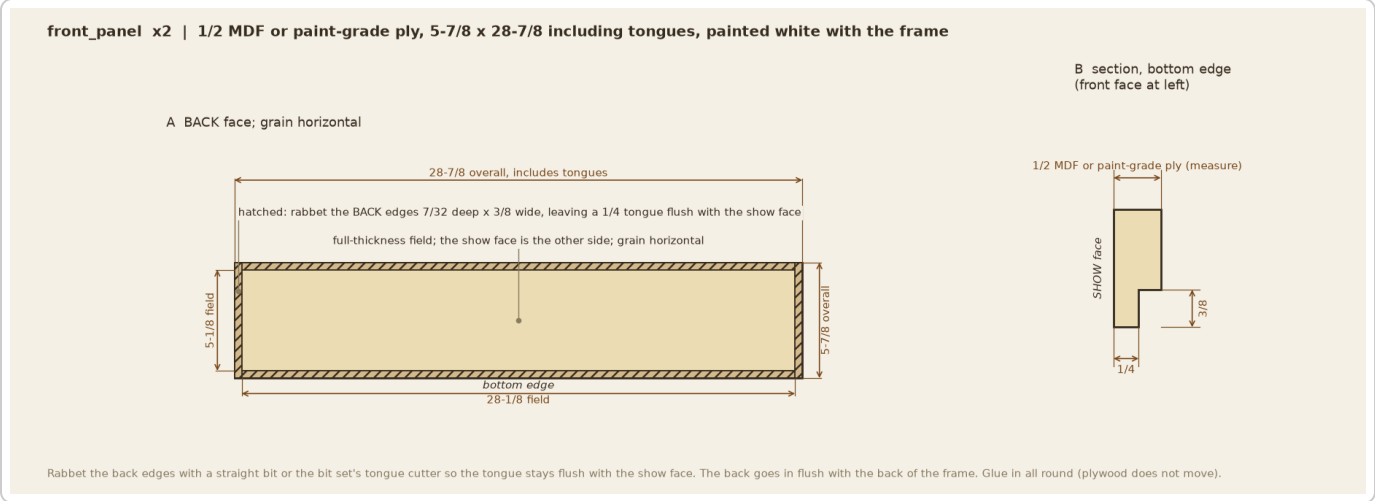
2 pieces · 2.4 sq ft face, no kerf or waste

2× front_panel

see drawing: not a plain rectangle

15/32 x 5-7/8 x 28-7/8

1/4 x 3/8 tongue on all four edges, flush with the show face (rabbet the BACK edges 7/32 deep); grain horizontal; size includes the tongues; back flush with the frame back, face 1/4 below the maple; glue in (plywood, no movement)



Drawer boxes: sides, fronts, backs and bottoms, 1/2 in 9-ply Baltic birch (measure the sheet)

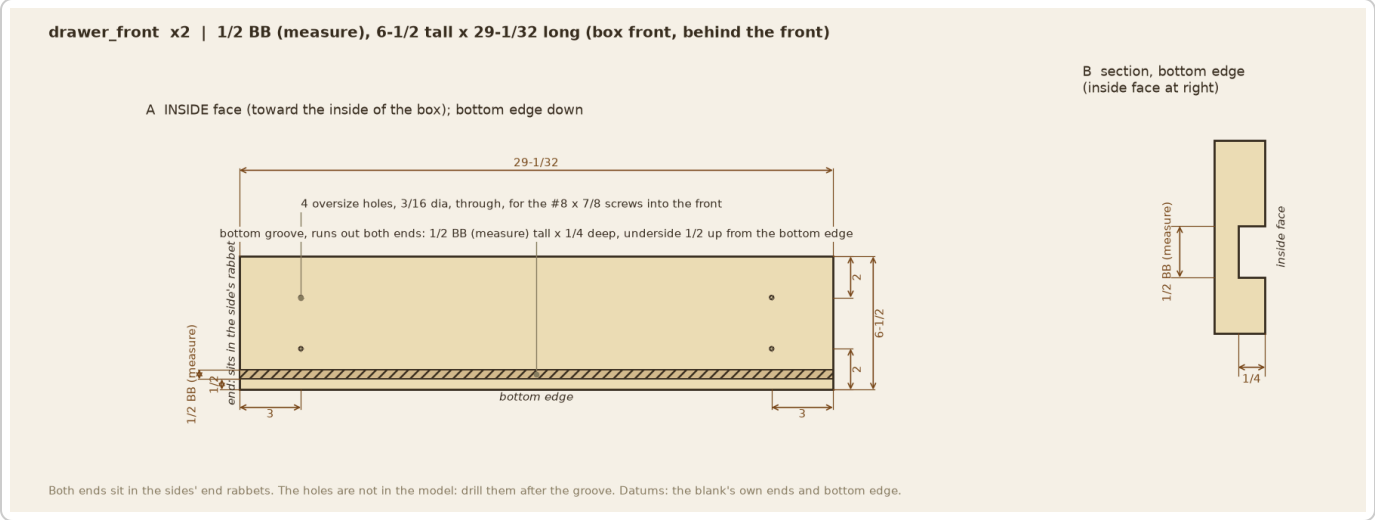
10 pieces · 17.2 sq ft face, no kerf or waste

2× drawer_front

see drawing: not a plain rectangle

15/32 x 6-1/2 x 29-1/32

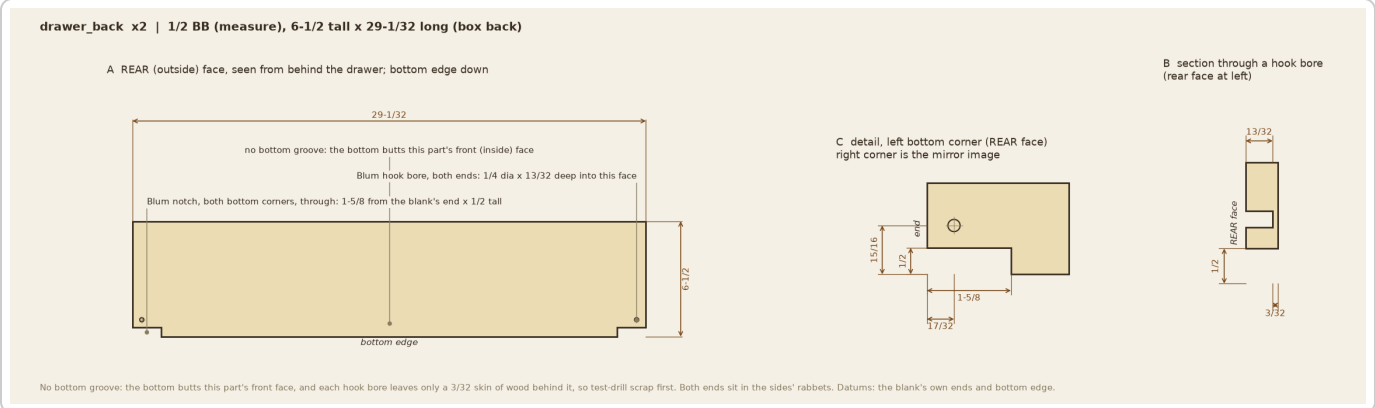
box front, sits in the side rabbets; 1/4 deep bottom groove to match the sides; the front screws to it from inside through 4 oversize holes: #8 x 7/8 in, pre-drilled, so the bite stays short of the face; the box front overlaps the top rail by 1-1/8 (best bite) and the panel field, but the bottom rail by only 1/4



2× drawer_back

15/32 x 6-1/2 x 29-1/32

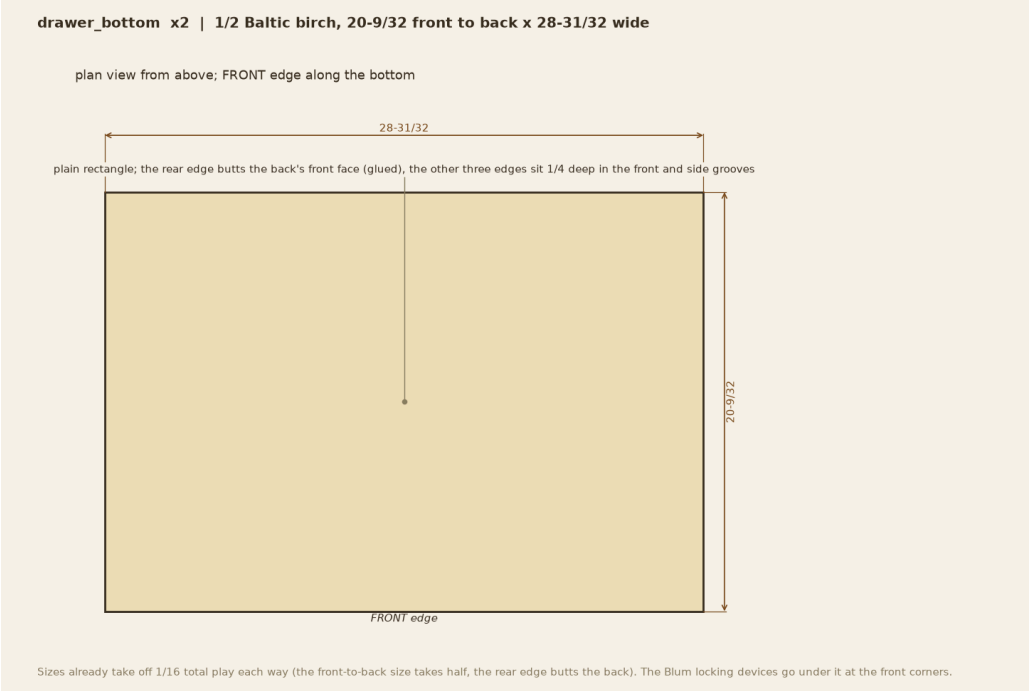
box back, sits in the side rabbets; no bottom groove (a groove would break into the hook bores), the bottom butts its front face; Blum notch 1-3/8 in from each side's inner face x 1/2 tall at both bottom corners; hook bores 1/4 dia x 13/32 deep into the rear face, centered 9/32 in from each side's inner face and 15/16 above the bottom edge (Blum 563H sheet)



2× drawer_bottom

15/32 x 20-9/32 x 28-31/32

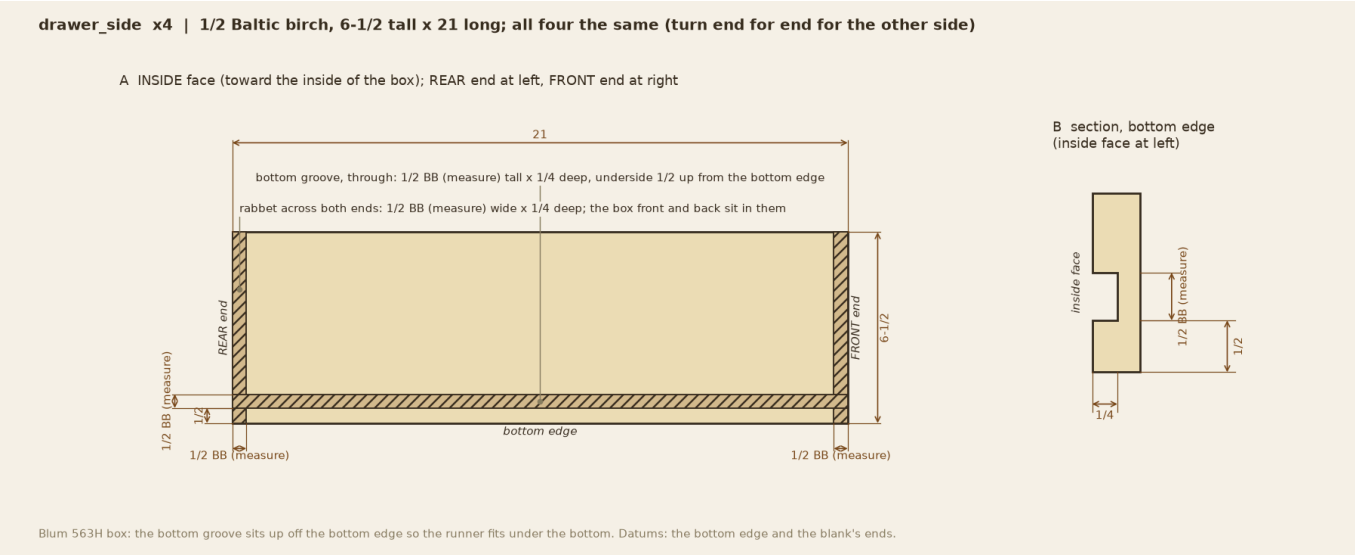
plain rectangle: 1/4 into the side and front grooves, butted to the back and glued to it; the Blum locking devices sit flush under it at the front corners (their screws go into the box front)



4× drawer_side see drawing: not a plain rectangle

15/32 x 6-1/2 x 21

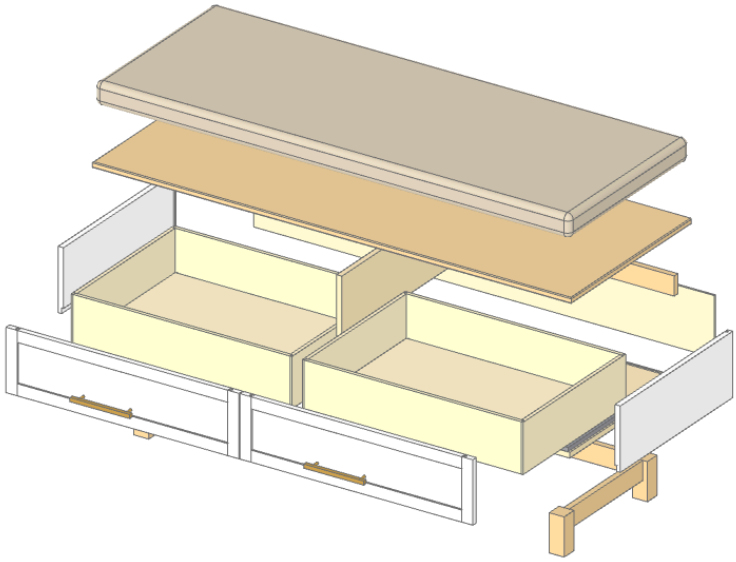
1/4 deep x 1/2 BB (measure) rabbet across both ends on the inside face; 1/4 deep x 1/2 BB (measure) bottom groove, underside 1/2 up from the bottom edge, through



Hardware from the design; not in the cut list

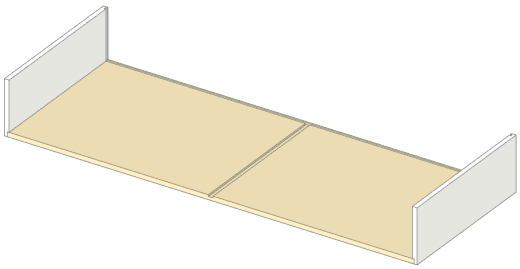
2 pair	**Blum TANDEM plus BLUMOTION 563H, 21 in (563H5330B)** one pair per drawer; 90 lb dynamic; runner front 3/32 behind the case front edge; no rear brackets in a frameless case
2 pair	**Blum locking devices T51.1901 R and L** front corners under the drawer bottom, screwed into the 1/2 box front with #6 x 1/2
2	**Brass bar pull, 8 in** centered on each front; break-away machine screws through the 1/2 panel plus the 1/2 box front
4	**Figure-8 tabletop fasteners** nailer to top; recess centered 1/4 in from the front face so it opens through
6	**Pocket screws #8 x 1-1/4 fine** ends and partition to the top, near the front
20	**Pocket screws #8 x 1-1/4 fine** base rails into the legs (2 per joint, 12) and up into the case bottom (10)
8	**Wood screws #8 x 2** down through the case bottom into the leg tops, two per leg, under the drawer boxes
8	**Wood screws #8 x 7/8** fronts to the box fronts, through oversize holes, pre-drilled
1 roll	**Iron-on maple edge banding, 13/16** the ends' front and bottom edges, about 62-1/2 in all, painted with them
1	**Bench cushion** about 63-1/2 x 24-1/2 x 3, boxed, the full depth of the top

Assembly 10 steps, in order



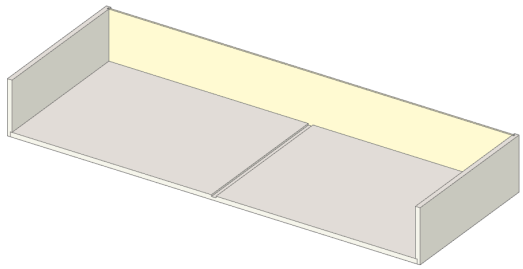
1. Ends and bottom

Glue the case bottom into the rabbets along the bottom of both ends, rabbets and grooves facing in, grooves at the rear, show faces out: a 62-1/2 x 23 case, 8-1/4 tall. Check for square.



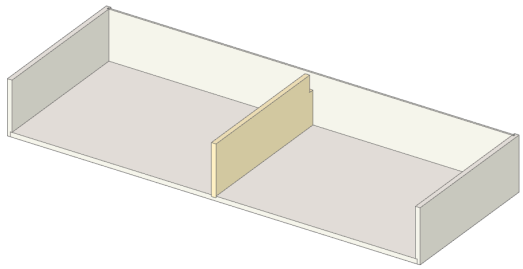
2. Back

Slide the 1/4 back down the end grooves until it seats in the bottom groove and glue it all round. With no wall screws it is what keeps the case square.



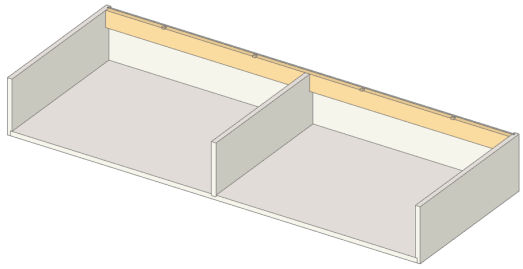
3. Partition

Drop the partition into the dado in the bottom with its notch at the top rear, front edge flush with the ends. Screw up through the bottom.



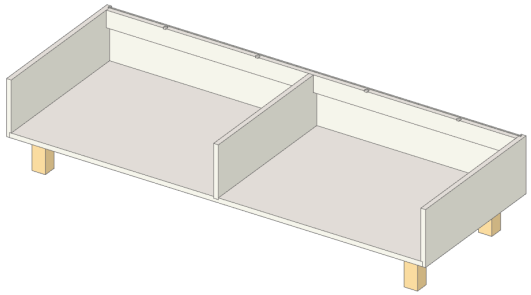
4. Nailer

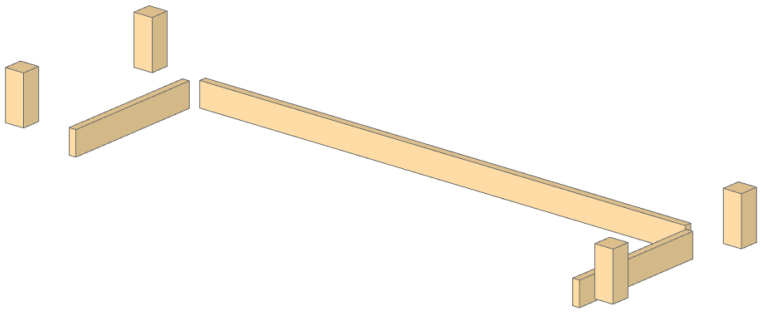
Slide the nailer in on edge through the partition notch, tight against the back and flush with the panel tops. Glue it to the back and screw through the ends into its ends. No wall screws: the piece is freestanding.



5. Base frame

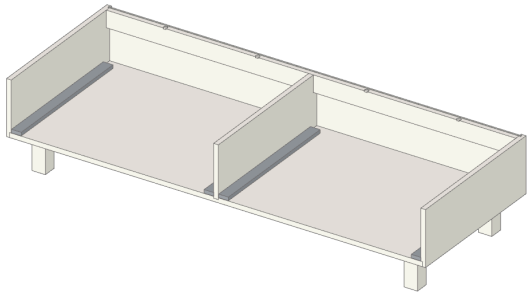
Glue and pocket-screw the two 2-1/2 in end rails between the front and back legs, centered on the legs with their top edges flush with the leg tops, and the back rail between the two back legs; no front rail. Turn the case over and set the frame on the case bottom: legs' outer faces 2-1/2 in from the case ends, front legs 2 in behind the case front edge, back legs 2 in inside the case back. Pocket screws up into the bottom, then two #8 x 2 down through the bottom into each leg top (they land under the drawer boxes). Turn it upright: 5 in of clear floor under the case.





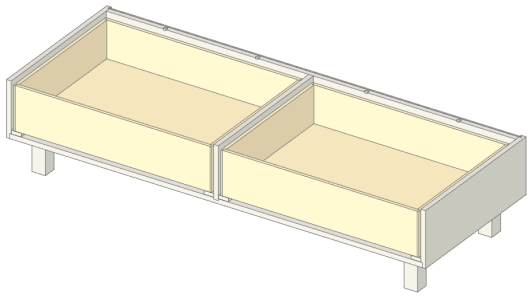
6. Slides

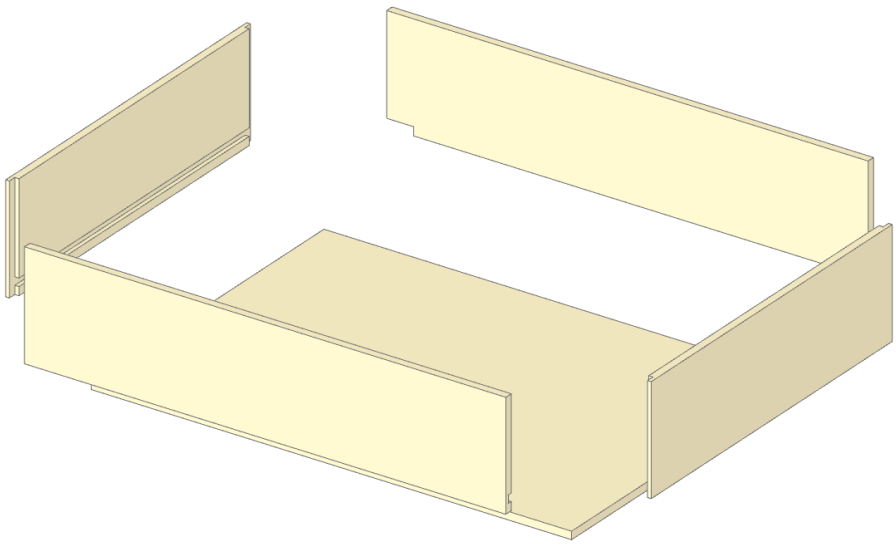
Screw the Blum cabinet members to the ends and the partition, sitting on the case bottom, fronts 3/32 behind the case front edge. Use the five marked holes per runner.



7. Drawer boxes

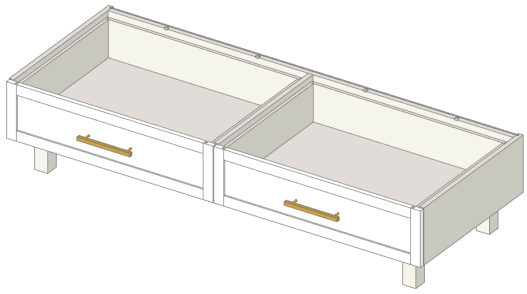
Glue each 1/2 birch box up around its bottom, 6-1/2 in tall: front and back in the side rabbets, bottom in the side and front grooves with its underside 1/2 above the side edges, butted to the back and glued to it. Notches and hook bores at the rear (drill scrap first: the bore leaves a thin skin). Clip the locking devices under the bottom at the front corners with #6 x 1/2 screws and set the boxes on the runners.

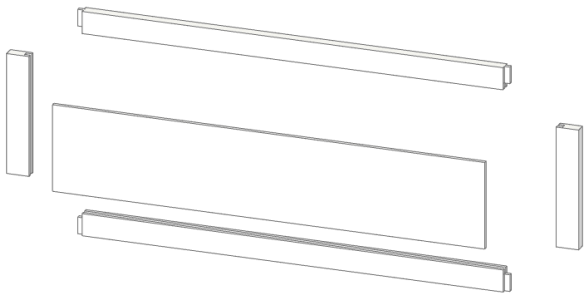




8. Fronts and pulls

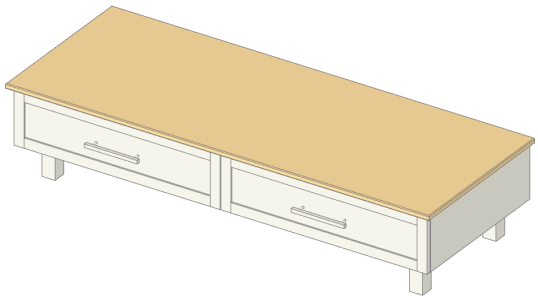
Glue up each 1/2 panel in its maple frame, back faces flush so the panel sits 1/4 below the maple (groove every piece first, then cut the tongues and tenons to fit). Prime and paint the fronts and the case ends white. Hang the fronts 1/16 in from each case end, 1/8 apart and 1/8 under the top, bottoms flush with the case underside, screwed from inside the box through oversize holes. Center the pulls.





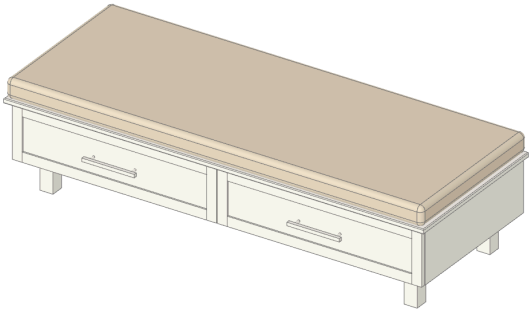
9. Top

Set the 64 x 24-1/2 top flush with the case back, overhanging 3/4 at each end and 25/32 at the front. Pocket screws up into it through the ends and partition near the front; figure-8s on the nailer at the back so it can move. Osmo Natural on the top, legs and rails.



10. Cushion

Cushion on. Slide the bench into the alcove with its back 1/2 off the wall. Done.



Generated 2026-10-26 from `projects/Freestanding-bench/cutlist.csv` and `images/parts/`, which `freestanding_bench.py` and `part_drawings.py` write on every export; every drawn cut is probed against the CAD solid before its sheet is written. The bench image at the top and the assembly pictures are screenshots of the CAD viewer (`hero_shot.py`, `assembly_shots.py`). Part names match the CAD viewer. The built-in version of this bench is at its own page.